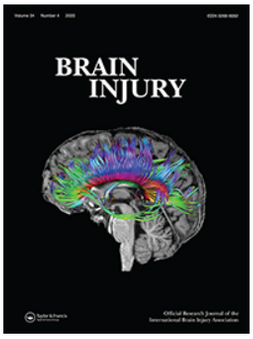

Brain Injury

ISSN: 0269-9052 (Print) 1362-301X (Online) Journal homepage: https://www.tandfonline.com/loi/ibij20

Chronic traumatic encephalopathy research viewed in the public domain: What makes headlines?

Daniel I. Wolfson, Andrew W. Kuhn, Zachary Y. Kerr, Benjamin L. Brett, Aaron M. Yengo-Kahn, Gary S. Solomon & Scott L. Zuckerman

To cite this article: Daniel I. Wolfson, Andrew W. Kuhn, Zachary Y. Kerr, Benjamin L. Brett, Aaron M. Yengo-Kahn, Gary S. Solomon & Scott L. Zuckerman (2020) Chronic traumatic encephalopathy research viewed in the public domain: What makes headlines?, Brain Injury, 34:4, 528-534, DOI: 10.1080/02699052.2020.1725843

To link to this article: https://doi.org/10.1080/02699052.2020.1725843

Published online: 17 Feb 2020.

Submit your article to this journal

Article views: 171

View related articles

View Crossmark data

Citing articles: 1 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ibij20

BRAIN INJURY
2020, VOL. 34, NO. 4, 528–534
https://doi.org/10.1080/02699052.2020.1725843

Taylor & Francis
Taylor & Francis Group

Check for updates

Chronic traumatic encephalopathy research viewed in the public domain: What makes headlines?

Daniel I. Wolfson[a,b], Andrew W. Kuhn[a], Zachary Y. Kerr[c,d,e], Benjamin L. Brett[a,f,g], Aaron M. Yengo-Kahn[a,b], Gary S. Solomon[a,b,h], and Scott L. Zuckerman[a,b]

[a]Vanderbilt Sports Concussion Center, Vanderbilt University Medical Center, Nashville, TN, USA; [b]Department of Neurological Surgery, Vanderbilt University Medical Center, Nashville, TN, USA; [c]Department of Exercise and Sport Science, University of North Carolina, Chapel Hill, NC, USA; [d]Matthew Gfeller Sport-Related Traumatic Brain Injury Research Center, University of North Carolina, Chapel Hill, NC, USA; [e]Injury Prevention Research Center, University of North Carolina, Chapel Hill, NC, USA; [f]Department of Neurology, Medical College of Wisconsin, Milwaukee, WI, USA; [g]Department of Neurological Surgery, Medical College of Wisconsin, Milwaukee, WI, USA; [h]Health and Safety Department, National Football League, New York, NY, USA

ABSTRACT

Objective: To determine chronic traumatic encephalopathy (CTE)-related publication characteristics associated with higher Altmetric scores.

Methods: A systematic review of the CTE literature was conducted using PubMed. Publications were coded for: journal impact factor (JIF); publication type (primary versus non-primary data collection); discussion of American football; contact sport-CTE association conclusion (yes versus no/neutral); and Altmetric score. Multivariable ordinal logistic regression identified predictors of higher Altmetric scores.

Results: Most of the 270 CTE-related publications did not include primary data collection (60%). The median Altmetric score was 12 (range = 0–3745). Higher Altmetric scores were associated with primary data collection [Odds ratio $(OR)_{Adjusted}$ = 2.29; 95% confidence interval (CI) = 1.35–3.89] and discussing American football ($OR_{Adjusted}$ = 2.11; 95%CI = 1.24–3.59). Among publications concluding contact sport-CTE associations, higher Altmetric scores were associated with higher JIF (3-point-JIF-increase $OR_{Adjusted}$ = 2.11; 95%CI = 1.24–3.59); however, the association of higher Altmetric scores with higher JIF was not found among neutral publications or those concluding no contact sport-CTE associations (3-point-JIF-increase $OR_{Adjusted}$ = 1.07; 95%CI = 0.94–1.22).

Conclusions: Most CTE-related publications (60%) did not involve primary data collection. Publication characteristics such as higher JIF and concluding contact sport-CTE associations were associated with higher Altmetric scores. It is important for the academic community to consider strategies to counter publication and promotion bias in the presentation of CTE literature.

ARTICLE HISTORY
Received 13 June 2019
Revised 26 January 2020
Accepted 1 February 2020

KEYWORDS
Chronic traumatic encephalopathy; sports; concussion; traumatic brain injury; social media

Introduction

In the context of sport-related concussion (SRC), the study of chronic traumatic encephalopathy (CTE) and neurodegenerative sequelae in former athletes has received international scientific and media attention. Based mostly on case reports from boxers and convenience samples of American football players, the public has largely accepted a causal link between contact sports and CTE (1–3). This may be due, in part, to heavily publicized accounts of CTE in prominent, retired National Football League (NFL) players (4). In a survey of 1,000 Americans, 83% thought that playing football "certainly" or "probably" causes long-term brain injuries; however, the scientific literature suggests that the overall mortality rate of retired professional football players is lower than or similar to the general population (5–8). It is possible the public has succumbed to the *availability cascade*, where individual, widely publicized examples readily available to the public receive disproportionate attention and supersede the findings of established empirical data (9,10). As several authors have asserted, it seems that many people have departed from the time-tested, measured, scientific approach to addressing a complex problem (11–17).

The public has a growing interest regarding CTE in sports. The aforementioned convenience sample of former football players suspected of having CTE was the most read Journal of American Medical Association (JAMA) article in 2017 (2,18). To fully understand the symbiotic relationship between science and the media, the evolving landscape of information dissemination and consumption should be taken into account. In-depth newspaper and magazine reporting of the results of scientific studies have taken a backseat to the internet news, blogs, soundbites, and tweets, and the CTE controversy seems to have been amplified by electronic headlines.

To measure the true influence of a scientific publication, alternative metrics, or *Altmetric Scores*, have been recently popularized (19). In addition to traditional citations, Altmetric scores utilize the total number of website views, downloads, tweets, mentions, and shares through news articles, blogs, internet searches, Facebook, and Twitter in order to quantify the impact of a scientific article into an aggregated

CONTACT Scott L. Zuckerman scott.zuckerman@vumc.org Department of Neurological Surgery, Vanderbilt University Medical Center, Medical Center North T-4224, Nashville, TN 37212

© 2020 Taylor & Francis Group, LLC

score (20). It has become apparent to the scientific community that to understand the societal impact of one's research, analyses must extend beyond the confines of scientific journals, traditional citation metrics, and PubMed (20,21). However, the intersection between reporting of results in scientific journals and media dissemination is poorly understood.

Due to the discordant public versus scientific opinions of the relationship between CTE and sports, we sought to improve our understanding of how this scientific information is disseminated and consumed. The objectives of this study were to: 1) report CTE publication characteristics in scientific journals and corresponding Altmetric scores, and 2) determine which objective publication characteristics predicted higher Altmetric scores.

Methods

Data search and study selection

A systematic literature review utilizing PubMed was performed on October 15th, 2018. The *a priori* objective was to identify studies devoted primarily to the study and/or discussion of CTE in sports. The following search terms were used: *"chronic traumatic encephalopathy"[tiab] AND (sport OR football OR soccer OR basketball OR hockey OR rugby OR wrestling OR boxing)*. CTE had to be mentioned either in the title or in the abstract, along with a mention of sports. All studies returned on the initial query were included. Since study inclusion was based purely on the search terms mentioned, no exclusion criteria were necessary. All publications were included, regardless of publication type. For each study, several publication characteristics were extracted that were based on two categories: 1) publication data, and 2) Altmetric scores.

Publication data

Publication data included: 1) year of publication; 2) journal of publication; 3) journal impact factor (JIF); 4) senior author institution (determined by corresponding author); 5) inclusion of CTE in title; 6) publication type; 7) population of interest; 8) mention of football; and 9) publication conclusion. JIF was determined by using the InCites Journal Citation Reports website, published by Clarivate (22). Senior author institution was recorded as the institution of the corresponding author. Publication type was classified according to the following categorical variables: case report; case series; case-control; cross-sectional; cohort; randomized controlled trial; narrative review; systematic review/meta-analysis; commentary/letter; and laboratory study. Any publication which was not a narrative review, systematic review/meta-analysis, or commentary/letter was considered primary data collection. For these studies with primary data collection, sample size information was recorded.

Mention of football and publication conclusion were subject to interpretation. These data were collected by four of the authors experienced in the research of SRC. To help ensure valid data acquisition for these variables, these four authors held meetings to establish reliably coding criteria. These meetings continued as coding ensued, with any questions or concerns about coding selection for specific publications openly discussed.

For mention of football, if the study focused on or mentioned American football in the majority of the abstract, title, and/or manuscript, this variable was coded as "Yes;" if the study focused primarily on a different sport, without mention of American football, this was coded as "No."

For the variable *publication conclusion*, the following operational definitions were employed with respect to the primary conclusions drawn from each publication:

- No contact sport-CTE association: Authors concluded that sports (e.g., participation and/or sport-related concussions/sub-concussive impacts) were not associated (cross-sectional or casual) with CTE, or that there currently lacked a strong body of evidence to make such conclusions.
- Neutral: Authors made no definitive conclusions about the association between contact sports and CTE.
- Contact sport-CTE association: Authors concluded that contact sports were associated with CTE, or that there currently existed a strong body of evidence to make such conclusions.

Altmetric scores

Altmetric scores were collected for each publication as a continuous variable based on a validated formula (23). The Altmetric Attention Score is an automatically calculated weighted count of the attention a research output has received based on three factors. First, "Volume" considers the number of mentions an article has, with an increasing score (one mention from each person per source) for more mentions. Second, "Source" provides different weights for the varying information sources through which the article is mentioned; for example, a newspaper article contributes more than a blog post or tweet. Final scores are calculated by summing the article's mentions, weighted based on the relative reach of each information source: news (weight of 8 per article), blogs (5), Wikipedia (3), Twitter (1), LinkedIn (0.5), Facebook (0.25), and YouTube (0.25). Third, "Author" allocates points based on analyses of the authors' social media usage and reach; this is used to place more value on organic public attention and minimize authors from marketing their own research or pushing their own agendas (24). In addition to the total Altmetric Score, we also collected additional scores provided for: news, blogs, twitter posts, Facebook posts, Wikipedia articles, Mendeley, and citations.

Statistical analysis

Data were analyzed with STATA version 14 (StataCorp LP, College Station, TX) and SAS version 9.4 (SAS Institute, Inc., Cary, NC). First, descriptive statistics were calculated to report CTE publication characteristics and Altmetric scores. Second, Wilcoxon-Rank Sum and chi-square tests compared distributions of publication characteristics between publications concluding and not concluding a contact sport-CTE association. Significance was determined *a priori* based on an alpha of <0.05. Third, we

examined the publication characteristics associated with Altmetric scores. Given the non-normal distribution and the wide range of Altmetric scores, three ordinal levels were used (<10, 10–50, >50). A multivariable ordinal logistic regression model estimated the odds of higher Altmetric score level with the following predictors: primary data collection (yes/no), inclusion of CTE in title (yes/no), discussion of American football (yes/no), and year (discrete variable). The model also included an interaction term between JIF (continuous variable) and contact sport-CTE association conclusion (yes versus no/neutral). Only publications with complete information for the predictor variables were included in the final model. ORs with 95% confidence intervals (CI) excluding 1.00 were deemed significant.

Results

Characteristics of publications

The PubMed search yielded 270 publications (Table 1). Years of publication ranged from 1987–2018, though 138 (51%) were published in 2016–2018, and 241 (89%) were published cumulatively since 2012 (Figure 1ab). The most common publication types were narrative reviews (42%), commentaries/letters (13%), and case series (9%). Most publications (60%) did not have primary data collection; of those publications with primary data collection, the median sample size was 36 [interquartile range (IQR) = 9–94; range = 1–3702). Thirty-five percent of publications involved studies with primary data collection in human samples.

Among all publications, 51% included CTE in the title and 67% discussed American football. Also, 39% concluded a contact sport-CTE association; 19% concluded no association and 41% remained neutral; two publications were excluded as these were unable to be accessed). JIF was available for all but 10 publications, with the median score being 3.67 (IQR = 2.03–5.87; range = 0.07–47.66). The top 5 most published institutions were Boston University (n = 37), University of Toronto (n = 11), Harvard/Brigham and Women's Hospital (n = 9), Vanderbilt University (n = 9), and University of California, Los Angeles (n = 7).

Table 1. Comparison of CTE publications concluding versus not concluding an association between contact sports and CTE.[a]

Characteristic	No/Neutral Contact Sport-CTE Association (n = 161)[b]	Yes Contact Sport-CTE Association (n = 107)	Total (n = 270)	p-value
Year of publication				0.11
Before 2000	4 (2%)	1 (1%)	6 (2%)	
2001 to 2009	4 (2%)	6 (6%)	10 (4%)	
2010 to 2012	10 (6%)	13 (12%)	23 (9%)	
2013 to 2015	63 (39%)	30 (28%)	93 (34%)	
2016 to 2018	80 (50%)	57 (53%)	138 (51%)	
Impact factor				0.11
Mean (SD)	5.18 (7.02)	6.49 (8.13)	5.65 (7.47)	
Median (range)	3.48 (0.06–47.66)	4.21 (0.14–47.66)	3.67 (0.07–47.67)	
Publication type, n (%)				<0.001*[c]
Non-Primary Data Collection	115 (71%)	44 (41%)	161 (60%)	
Narrative review	77 (48%)	33 (31%)	111 (41%)	
Commentary/letter	27 (17%)	8 (7%)	36 (13%)	
Systematic review	11 (7%)	3 (3%)	14 (5%)	
Primary Data Collection	46 (29%)	63 (59%)	109 (40%)	
Cohort	7 (4%)	10 (9%)	17 (6%)	
Case-control	7 (4%)	14 (13%)	21 (8%)	
Cross-sectional	13 (8%)	5 (5%)	18 (7%)	
Case series	10 (6%)	13 (12%)	23 (9%)	
Case report	3 (2%)	14 (13%)	17 (6%)	
Laboratory	6 (4%)	7 (7%)	13 (5%)	
Sample size (primary data collection studies only)				0.04*
Mean (SD)	305 (831)	107 (277)	182 (560)	
Median (range)	40 (1–3702)	23 (1–1721)	36 (1–3702)	
CTE in title, n (%)	67 (42%)	69 (65%)	137 (51%)	<0.001*
Football discussed, n (%)	100 (62%)	79 (74%)	180 (67%)	0.046*
Altmetric score				0.002*
Mean (SD)	29 (57)	137 (416)	71 (270)	
Median (range)	10 (0–577)	19 (0–3745)	12 (0–3745)	
Outlet, median (range)				
General and social media				
News	0 (0–73)	1 (0–276)	0 (0–276)	0.004*
Blogs	0 (0–6)	0 (0–41)	0 (0–41)	<0.001*
Tweets	5 (0–138)	7 (0–1570)	0 (0–1570)	0.13
Facebook	0 (0–40)	0 (0–102)	0 (0–102)	0.12
Wiki	0 (0–2)	0 (0–4)	0 (0–4)	0.005*
Research-based outlets				
Mendeley	35 (0–536)	48 (0–835)	39 (0–835)	0.01*
Citations	7 (0–254)	13 (0–1074)	9 (0–1074)	0.002*

CTE, chronic traumatic encephalopathy.

* denotes statistical significance ($p < 0.05$) from Wilcoxon-Rank Sum test for discrete/continuous variables and chi-square/Fisher's Exact test for categorical variables.

[a]Two publications were excluded from conclusion type comparison because researchers were unable to access abstracts or manuscripts in order to evaluate publication conclusions.

[b]No/neutral included 110 publications positing no association and 51 remaining neutral.

[c]Chi-square test compared distributions of primary and non-primary data collection publications by conclusion type.

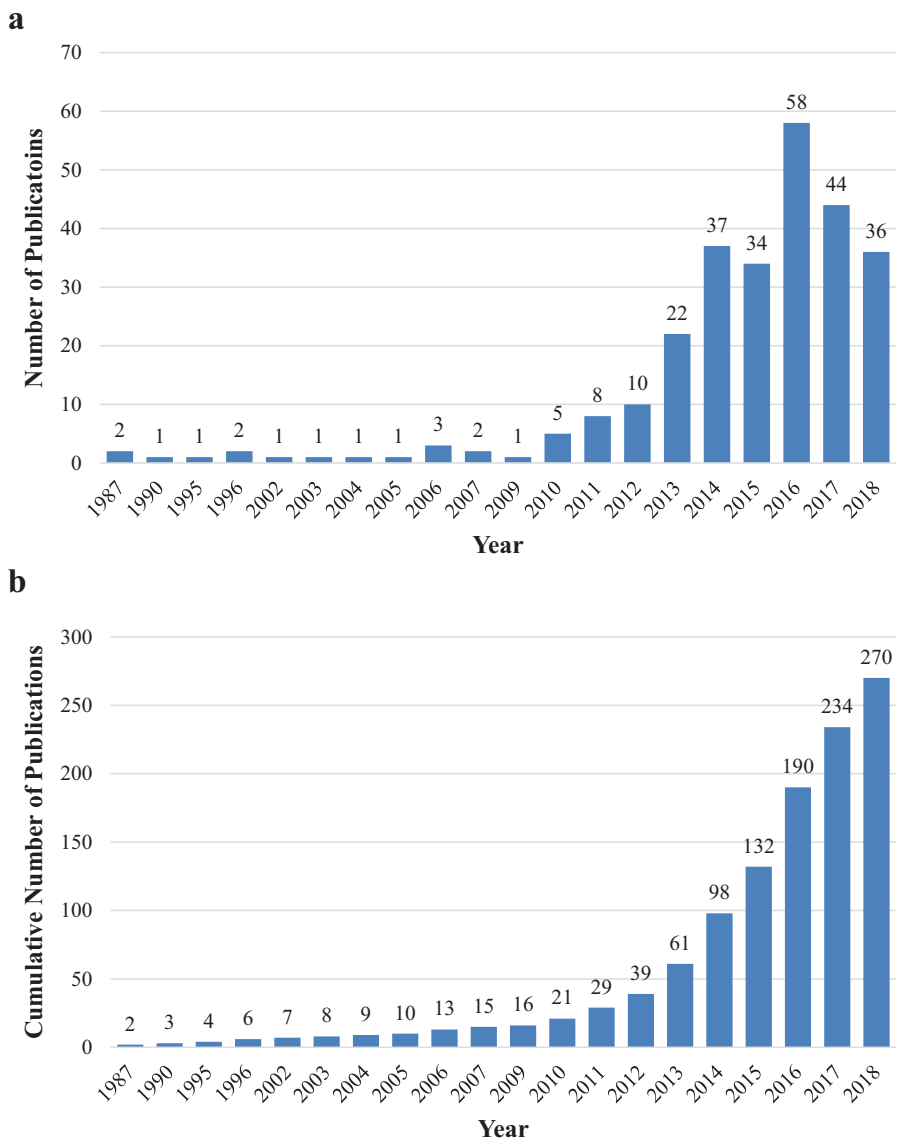

Figure 1. Number of CTE publications, per year (a) and cumulatively (b).

The median Altmetric score was 12 (IQR = 2–45). However, the range of Altmetric scores ranged from 0 to 3745; 43.7% (n = 118) of publications had Altmetric scores of <10, 34.4% (n = 93) had scores of 10–50, and 21.9% (n = 59) had scores >50 (Figure 2). Twitter was one of the most used general/social media outlet (median = 5; IQR = 2–21; range = 0–1570).

Comparison based on study conclusion of contact sport-CTE association

Publications concluding the presence of a contact sport-CTE association were more likely to be primary data collection studies ($p < .001$), include CTE in the title ($p < .001$), discuss American football ($p = .046$), and have higher Altmetric scores ($p = .002$) (Table 1). In addition, among primary data collection studies, publications concluding the presence of a contact sport-CTE association were more likely to have a smaller sample size ($p = .04$). In particular, 25% of publications concluding the presence of a contact sport-CTE association were case series and case reports, compared to 8% for publications not concluding the presence of a contact sport-CTE association.

Multivariable ordinal logistic regression model predicting higher altmetric scores

In the multivariable model, the proportional odds assumption was met ($p = .96$) (Table 2). Controlling for all covariates, the odds of higher Altmetric score level were higher for use of primary data collection ($OR_{Adjusted} = 2.29$; 95%CI = 1.35–3.89), inclusion of CTE in title ($OR_{Adjusted} = 2.35$; 95%CI = 1.41–3.92), and mention of American football ($OR_{Adjusted} = 2.11$; 95%CI = 1.24–3.59). Also, when examining the interaction between JIF and study conclusion, among publications concluding a contact sport-CTE association, odds increased with a higher JIF (3-point-increase $OR_{Adjusted} = 2.11$; 95%CI = 1.24–3.59); however, among publications not concluding a contact sport-CTE association, JIF was not associated with a change in odds (3-point-increase $OR_{Adjusted} = 1.07$; 95% CI = 0.94–1.22).

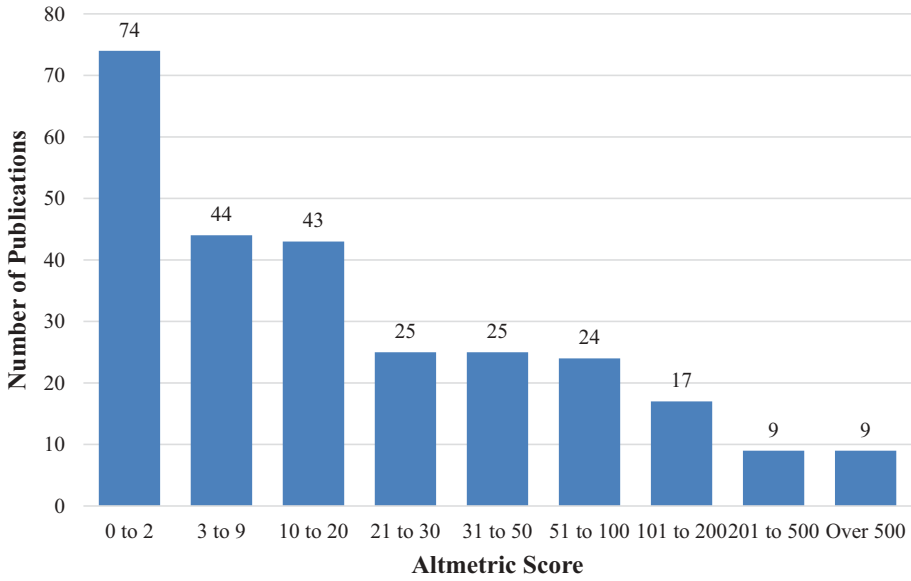

Figure 2. Distribution of Altmetric scores of CTE publications.

Table 2. Multivariable ordinal logistic regression[a] examining factors associated with higher Altmetric score levels among CTE publications.

Characteristic	Adjusted odds ratio (95%CI)
Primary data collection	
Yes	2.29 (1.35–3.89)*
No	1.00
CTE in title	
Yes	2.35 (1.41–3.92)*
No	1.00
Football discussed	
Yes	2.11 (1.24–3.59)*
No	1.00
Year (1-year increase)	1.06 (0.99–1.13)
Journal impact factor (3-point increase)	
Among studies concluding contact sport-CTE association	2.11 (1.24–3.59)*
Among studies not concluding contact sport-CTE association	1.07 (0.94–1.22)

CTE, chronic traumatic encephalopathy.
*denotes statistical significance (i.e., odds ratio 95%CI did not include 1.00)
[a]Given the non-normal distribution and the wide range of Altmetric scores, three ordinal levels were used (<10, 10–50, >50).

Discussion

The current study sought to examine the consumption and dissemination of scientific publications focused on CTE in sports, as evaluated by Altmetric scores. Overall, three in five publications did not contain primary data. Higher Altmetric scores were associated with the use of primary data collection, the inclusion of CTE in the title, discussion of American football, and JIF. However, the association between higher Altmetric score and JIF did not hold for neutral publications or those concluding no contact sports-CTE link. The results of the current study shed light on a controversial public health dilemma and spark debate for future study.

Reviewing the selected characteristics of all included publications reveals a lack of empirical data on CTE in humans. Over half of all publications were review papers without primary data and only 35% of publications involved studies with primary data collection in human samples. Also, the large majority of publications were published within the past decade, with over half in 2016–2018. Further, there was not a consensus in the association of contact sports and CTE with our estimations, noting that 39% supported such a conclusion, 19% did not, and 41% remained neutral, with many noting the need for more exploration. Overall, these results highlight that regarding CTE in sports, there is more discussion and review of existing data than the creation of new data, a finding likely related to the limitations of only antemortem detection/classification and poor availability of CTE-specific autopsy data. Though the concept of CTE is old, the comprehensive study of modern CTE in sports is relatively young, and it may be appropriate to soften conclusions and scale back apparent definitive claims given the limited empirical data.

Altmetric scores allow us to draw conclusions regarding the public impact of CTE research. The median Altmetric score of the top 25 CTE publications was 377, which is comparably high relative to the median Altmetric scores of the top 25 articles in the fields of Plastic Surgery (score = 23), Neurosurgery (score = 184), and General Surgery (score = 22, from top 10) (20,24–26). In comparing articles that did or did not conclude an association between contact sports and CTE, Altmetric scores were higher in those that supported an association. This raises concern for *publication bias*, where studies with positive associations are more likely to be published and cited (27). Further, there may also be a *promotion bias*, in which such work is less disseminated through traditional new sources and consumer-based news outlets may be more excited by "positive" results than "negative" ones. It is for this reason that several established researchers within the field have supported this notion (11–17). Stewart and 60 colleagues (all engaged in TBI research and practice) recently published a joint statement emphasizing the many remaining uncertainties must be highlighted when communicating about

CTE. If not done, wrong information may be disseminated – clinicians must first "do no harm" when communicating on CTE (17).

In further exploration of the association between study conclusion and Altmetric scores, an interaction between study conclusion and JIF existed. Higher JIF scores were associated with higher Altmetric scores, but only when the publication concluded a contact sports-CTE association. This relationship is consistent with the intuitive notion that higher impact factor journals are higher profile and lead to more public attention for their articles, as a prior orthopedic study has shown(28). Interestingly, however, the modification of this association between JIF and Altmetric score by study conclusion suggests that *publication bias* may not be the only factor at play. It appears that the public may be more likely to read and spread information about studies finding contact sport-CTE associations, and less likely to consume and disseminate publications that do not find such an association, or remain neutral. The degree to which this phenomenon is unique to CTE is unclear. However, such discussion of studies is also increased with not only CTE in the title, but also discussion of American football, which is the primary context in which the association of contact sports and CTE has been publicly discussed.

As researchers produce new knowledge, journals have an active role in dissemination as well. Journals can combat *promotion bias* by equally promoting studies with negative findings through twitter and other social media routes (29). For example, a study that found accelerated tauopathy in the retinas of mice exposed to mild traumatic brain injury received an Altmetric score of 21 while a prospective, controlled, and blinded study of 20 amateur boxers who showed no evidence of neuropsychologic deterioration over a period of nine years received an Altmetric score of 0 (30,31).

In addition, these results can also be used to improve communication between the scientific and public communities, ultimately shaping the scientific narrative in a more responsible manner. First, Kuhn et al. have suggested encouraging a synergistic relationship between the scientific and media communities (32). Forums for collaboration between both parties should be pursued that allow discussion and cordial discourse, followed by dissemination of such proceedings. Moreover, as over one-quarter of all CTE publications come from only five institutions, it appears that the current CTE research ecosystem may lack the diversity of thought that is the lifeblood of scientific progress. Institutions not only in the U.S., but across the world, must be encouraged to enter the discussion on CTE in sports and share their different perspectives.

In regards to the advent of social media, active promotion of scientific discussion needs to be inclusive of the growing network of physicians and researchers on social media who possess a direct outlet to the public at large, and not only the academic journals and institutions (33). However, such use is not without risks, as the lack of agreed-upon standards of practice on medical social media may also lead to misinformation, conflicting advice, and lapses in professionalism, particularly with an emotional topic such as CTE. We also highly encourage discussion on social media to not replace the need for scientific data collection and dialogue within academic journals.

Limitations

The current study has limitations that must be considered in the context of our study conclusions. First, only one database was used – PubMed – and CTE publications in other, international databases may have been missed. However, given the high prevalence of research investigating CTE in American Football, the effects of this are likely minimal. Second, our study limited the number of scientific variables included (e.g., year of publication, JIF, publication type, etc.) and it is possible that the inclusion of other scientific variables could affect our results and study conclusions. However, these scientific variables were selected *a priori* as we believed they would be the most important to examine. Third, there is inherent nuance in determining whether or not a publication supports a conclusion of an association between contact sports and CTE. Although the study's coders were trained to ensure comparability, and consistent meetings were held to identify and resolve any coding issues, it is nonetheless possible that our coding may have been susceptible to misclassification. Finally, Altmetric scores are constantly changing and naturally increase over time as global social media usage expands, likely resulting in higher scores for more recent publications. We sought to control this confounding through including year of publication in our multivariable model. Nonetheless, replication of our study may be difficult as Altmetric scores may increase over time. We encourage continued exploration of this area of study.

Conclusions

The discussion of CTE in publications has increased in the past decade, with most research occurring in the last five years. However, the majority of these publications did not involve primary data. Certain characteristics of CTE-related publications were associated with higher Altmetric scores, including JIF and study conclusion. It is important for the academic community, inclusive of researchers and the academic journals, to consider strategies to counter publication and promotion bias in the presentation of CTE literature. Further co-education between scientific and media communities is also needed to clarify current knowledge gaps in CTE literature.

Disclosure of Interest

Gary S. Solomon is a consultant for the Nashville Predators, Tennessee Titans, and the athletic departments of Tennessee Tech University and the University of Tennessee, with the fees paid to his institution. He is also a consultant to the National Football League Department of Health and Safety. The remaining authors declare that the article content was composed in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

1. Omalu B, Bailes J, Hamilton RL, Kamboh MI, Hammers J, Case M, Fitzsimmons R. 2011. Emerging histomorphologic phenotypes

of chronic traumatic encephalopathy in American athletes. Neurosurgery. 69(1):173–83. doi:10.1227/NEU.0b013e318212bc7b
2. Mez J, Daneshvar DH, Kiernan PT, Abdolmohammadi BA, Alvarez VE, Huber BR, Alosco ML, Solomon TM, Nowinski CJ, McHale L, et al. Clinicopathological evaluation of chronic traumatic encephalopathy in players of American football. JAMA. 2017;318(4):360. doi:10.1001/jama.2017.8334.
3. McCrory P, Zazryn T, Cameron P. 2007. The evidence for chronic traumatic encephalopathy in boxing. Sport Med. 37(6):467–76. doi:10.2165/00007256-200737060-00001
4. Omalu BI, DeKosky ST, Minster RL, Kamboh MI, Hamilton RL, Wecht CH. 2005. Chronic traumatic encephalopathy in a National Football League player. Neurosurgery. 57(1):128–34. doi:10.1227/01.NEU.0000163407.92769.ED
5. Poll: Three-Quarters of Football Fans See Head Injuries as Major Problem. Lowell (MA): UMass Lowell; 2017 Sep 6 [accessed 2018 Nov 29]. https://www.uml.edu/News/press-releases/2017/NFLPoll09062017.aspx.
6. Baron SL, Hein MJ, Lehman E, Gersic CM. 2012. Body mass index, playing position, race, and the cardiovascular mortality of retired professional football players. Am J Cardiol. 109(6):889–96. doi:10.1016/j.amjcard.2011.10.050
7. Lincoln AE, Vogel RA, Allen TW, Dunn RE, Alexander K, Kaufman ND, Tucker AM. 2018. Risk and causes of death among former National Football League players (1986–2012). Med Sci Sports Exerc. 50(3):486–93. doi:10.1249/MSS.0000000000001466
8. Venkataramani AS, Gandhavadi M, Jena AB. 2018. Association between playing American football in the National Football League and long-term mortality. JAMA. 319(8):800–06. doi:10.1001/jama.2018.0140
9. Barr WB. 2013. An evidence based approach to sports concussion: confronting the availability cascade. Neuropsychol Rev. 23(4):271–72. doi:10.1007/s11065-013-9244-3
10. Solomon GS, Sills A. 2014. Chronic traumatic encephalopathy and the availability cascade. Phys Sportsmed. 42(3):26–31. doi:10.3810/psm.2014.09.2072
11. Asken BM, Sullan MJ, DeKosky ST, Jaffee MS, Bauer RM. 2017. Research gaps and controversies in chronic traumatic encephalopathy. JAMA Neurol. 74(10):1255. doi:10.1001/jamaneurol.2017.2396
12. Randolph C. 2014. Is chronic traumatic encephalopathy a real disease?. Curr Sports Med Rep. 13(1):33–37. doi:10.1097/OPX.0000000000000170
13. Solomon G. 2018. Chronic traumatic encephalopathy in sports: a historical and narrative review. Dev Neuropsychol. 43(4):279–311. doi:10.1080/87565641.2018.1447575
14. Iverson GL, Gardner AJ, McCrory P, Zafonte R, Castellani RJ. A critical review of chronic traumatic encephalopathy. Neurosci Biobehav Rev. 2015;56:276–93. doi:10.1016/j.neubiorev.2015.05.008.
15. Castellani RJ, Perry G, Iverson GL. 2015. Chronic effects of mild neurotrauma: putting the cart before the horse?. J Neuropathol Exp Neurol. 74(6):493–99. doi:10.1097/NEN.0000000000000193
16. Davis GA, Castellani RJ, McCrory P. 2015. Neurodegeneration and sport. Neurosurgery. 76(6):643–56. doi:10.1227/NEU.0000000000000722
17. Stewart W, Allinson K, Al-Sarraj S, Bachmeier C, Barlow K, Belli A, Burns MP, Carson A, Crawford F, Dams-O'Connor K, et al. Primum non nocere: a call for balance when reporting on CTE. Lancet Neurol. 2019;18(3):231–33. doi:10.1016/S1474-4422(19)30020-1.
18. Abbasi J. 2018. JAMA Network articles of the year. JAMA. 320(21):2188. doi:10.1001/jama.2018.17934
19. Serghiou S, Ioannidis JPA. 2018. Altmetric scores, citations, and publication of studies posted as preprints. JAMA. 319(4):402. doi:10.1001/jama.2017.21168
20. Wang J, Alotaibi NM, Ibrahim GM, Kulkarni AV, Lozano AM. The spectrum of Altmetrics in neurosurgery: the Top 100 "trending" articles in neurosurgical journals. World Neurosurg. 2017;103:883–895.e1. doi:10.1016/J.WNEU.2017.04.157.
21. Campbell EG, Donelan K, DesRoches C, Roman A, Bolcic-Jankovic D. 2012. The patient-doctor relationship and online social networks: results of a national survey. J Gen Intern Med. 27(4):403–403. doi:10.1007/s11606-012-1989-5
22. Welcome to Journal Citation Reports. Clarivate Analytics; [accessed 2018 Nov 26]. https://jcr.clarivate.com/JCRLandingPageAction.action.
23. How is the Altmetric Attention Score calculated? Altmetric Support; 2018 Apr 5 [accessed 2018 Nov 26]. https://help.altmetric.com/support/solutions/articles/6000060969-how-is-the-altmetric-attention-score-calculated-.
24. Dardas LA, Woodward A, Scott J, Xu H, Sawair FA. Measuring the social impact of nursing research: an insight into altmetrics. J Adv Nurs. January 2019. doi:10.1111/jan.13921.
25. Ruan QZ, Chen AD, Cohen JB, Singhal D, Lin SJ, Lee BT. 2018. Alternative metrics of scholarly output. Plast Reconstr Surg. 141(3):801–09. doi:10.1097/PRS.0000000000004128
26. Powell AGMT, Bevan V, Brown C, Lewis WG. 2018. Altmetric versus bibliometric perspective regarding publication impact and force. World J Surg. 42(9):2745–56. doi:10.1007/s00268-018-4579-9
27. Dumas-Mallet E, Smith A, Boraud T, Gonon F. 2017. Poor replication validity of biomedical association studies reported by newspapers. PLoS One. 12(2):e0172650. doi:10.1371/journal.pone.0172650
28. Hughes H, Hughes A, Murphy C. 2017. The use of Twitter by the trauma and orthopaedic surgery journals: twitter activity, impact factor, and alternative metrics. Cureus. 9(12):e1931. doi:10.7759/cureus.1931
29. Thoma B, Murray H, Huang SYM, Milne WK, Martin LJ, Bond CM, Mohindra R, Chin A, Yeh CH, Sanderson WB, et al. The impact of social media promotion with infographics and podcasts on research dissemination and readership. CJEM. 2018;20(2):300–06. doi:10.1017/cem.2017.394.
30. Xu L, Ryu J, Nguyen JV, Arena J, Rha E, Vranis P, Hitt D, Marsh-Armstrong N, Koliatsos VE. Evidence for accelerated tauopathy in the retina of transgenic P301S tau mice exposed to repetitive mild traumatic brain injury. Exp Neurol. 2015;273:168–76. doi:10.1016/j.expneurol.2015.08.014.
31. Porter MD. 2003. A 9-year controlled prospective neuropsychologic assessment of amateur boxing. Clin J Sport Med. 13(6):339–52. doi:10.1097/00042752-200311000-00002
32. Kuhn AW, Yengo-Kahn AM, Kerr ZY, Zuckerman SL. 2017. Sports concussion research, chronic traumatic encephalopathy and the media: repairing the disconnect. Br J Sports Med. 51(24):1732–33. doi:10.1136/bjsports-2016-096508
33. Pershad Y, Hangge PT, Albadawi H, Oklu R. Social medicine: twitter in healthcare. J Clin Med. 2018;7(6). doi:10.3390/jcm7060121